# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

## CLASS 9 — SCIENCE UNIT TEST SET C

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

**Time: 2 Hours** Name: _____ Roll No.: _____ **Maximum Marks: 50**

**General Instructions:**

1. All questions are compulsory.
2. Q. 1–12 are Objective Type (1 mark each).
3. Q. 13–17 are Very Short Answer Type (2 marks each).
4. Q. 18–23 are Short Answer Type (3 marks each).
5. Q. 24–25 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 22 to 25.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

## SECTION — A (Objective Type Questions) 12 × 1 = 12

*Choose the correct option and write it in your answer book.*

**1.** The SI unit of acceleration is: 1

**(a)** m s² **(b)** km h⁻¹ **(c)** m s⁻² **(d)** m s⁻¹

**2.** A child on a merry-go-round of radius 7 m completes one full revolution (π = 22/7). The distance travelled and the displacement are: 1

**(a)** 44 m and 14 m **(b)** 44 m and 0 **(c)** 0 and 44 m **(d)** 22 m and 7 m

**3.** A speed of 72 km h⁻¹ in SI units is: 1

**(a)** 20 m s⁻¹ **(b)** 24 m s⁻¹ **(c)** 259.2 m s⁻¹ **(d)** 40 m s⁻¹

**4.** Two teams in a tug of war pull with 500 N each and the rope does not move. The forces on the rope are: 1

**(a)** unbalanced **(b)** zero on each team
**(c)** action–reaction on the rope **(d)** balanced

**5.** 1 N is the force that gives an acceleration of 1 m s⁻² to a mass of: 1

**(a)** 1 g **(b)** 9.8 kg **(c)** 10 kg **(d)** 1 kg

**6.** Two forces of 8 N and 8 N act on a block along the same straight line in opposite directions. The magnitude of the net force is: 1

**(a)** 0 N **(b)** 16 N **(c)** 64 N **(d)** 8 N

**7.** The speed of a child reaching the bottom of a smooth slide of height h depends on: 1

**(a)** h only (and g) **(b)** the shape of the slide
**(c)** the mass of the child **(d)** the time taken

**8.** Work done by a goalkeeper in stopping a ball is: 1

**(a)** zero **(b)** negative **(c)** infinite **(d)** positive

**9.** A machine lifts a load of 500 N by applying an effort of 125 N. Its mechanical advantage is: 1

**(a)** 5 **(b)** 8 **(c)** 4 **(d)** 0.25

**10.** Sound of frequency 500 Hz travels at 1500 m s⁻¹. Its wavelength is: 1

**(a)** 6 m **(b)** 750000 m **(c)** 3 m **(d)** 0.33 m

**11.** Persistence of sound due to multiple reflections is called: 1

**(a)** reverberation **(b)** resonance **(c)** pitch **(d)** echo

**12.** Sound travels fastest in: 1

**(a)** gases **(b)** solids **(c)** vacuum **(d)** liquids

## SECTION — B (Very Short Answer Type) — 5 × 2 = 10

*Answer in about 10–15 words each.*

**13.** Convert 90 km $h^{-1}$ into m $s^{-1}$. 2

**14.** Find the force required to accelerate a 1500 kg car at 2 m $s^{-2}$. 2

**15.** Define 1 joule of work. 2

**16.** Why is sound called a longitudinal wave? 2

**17.** Define wavelength, frequency and time period of a sound wave. 2

## SECTION — C (Short Answer Type) — 6 × 3 = 18

*Answer in about 30–40 words each.*

**18.** Derive $v = u + at$ using the definition of acceleration. 3

**19.** A body of mass 10 kg is acted upon by a force of 50 N. Find the acceleration and the distance covered in 4 s from rest. 3

**20.** A child slides down a smooth slide of height 2.5 m. Find the speed at the bottom ($g = 10$). 3

**21.** Derive $v = \nu \lambda$. 3

**22.** A 25 kg block is pushed with 55 N against a maximum friction of 50 N. Find the acceleration and displacement in 2 s. 3

OR

Does a body moving with constant velocity have a net force on it? Explain with the example of a box pushed on a floor. 3

**23.** A goalkeeper stops a ball with a force of 200 N, her hands moving back 15 cm. Find the work done on the ball. 3

OR

Derive the expression for gravitational potential energy. 3

## SECTION — D (Long Answer Type) — 2 × 5 = 10

*Answer in about 100–150 words each.*

**24.** **Case study:** A sports car of mass 1500 kg moves east: 0 to 10 m $s^{-1}$ in 5 s, then constant for 5 s, then to rest in 5 s. 5

(i) Find the force in the first 5 s.
(ii) Find the net force in the middle interval.
(iii) Find the force during braking.
(iv) Find the total distance.

OR

(i) State Newton's third law. 5
(ii) Explain with the example of a canoe.
(iii) Why do the forces not cancel?

**25.** **Case study:** Human hearing spans about 20 Hz to 20 kHz; take $v = 344$ m $s^{-1}$. 5

(i) Find the wavelength at 20 Hz.
(ii) Find the wavelength at 20 kHz.
(iii) Which is infrasonic and which ultrasonic?
(iv) Name one animal that detects each.

OR

(i) Explain the propagation of sound as compressions and rarefactions. 5
(ii) Is the medium carried along?
(iii) Distinguish longitudinal and transverse waves.

## END OF QUESTION PAPER — Total: 50 Marks